AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer with a background in Full-Stack Software Engineering. Highly motivated to apply my expertise in Machine Learning, Deep Learning, and Mathematical Modeling to the research challenges at Delft University of Technology. Experienced in Industrial Digital Twins, Healthcare, Financial Trading, and Computer Vision.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL
ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX
AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM
Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization
Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS
Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy
Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra
Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern) Jan 2026 – Jul 2026
Sonatrach (Industrial Surveillance Division) — Algeria
Architected an AI-native Digital Twin for real-time monitoring of Gas Turbines.

- Architected a 3-tier Dual-Track framework decoupling safety-critical Edge models from Cloud analytics.
- Developed **TCNEdge** achieving **0.049ms INT8 ONNX C++ inference** for emergency shutdown.
- Engineered **MambaTS** with a missingness-aware gate to maintain robustness during sensor dropouts.
- Implemented **KAN-PCNN** and **ST-KDFormer** to ensure strict thermodynamic consistency at scale.

Network Engineering Intern Sep 2024
Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA) 2021–2026
Engineering and Master's Degree in Computer Science
Specialization: Artificial Intelligence & Data Science

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

ACADEMIC THESES

State Engineering Degree Thesis:
Physics-Informed Explainable Digital Twin for Industrial Gas Turbines: A Dual-Track Cloud–Edge Deep Learning Framework with Koopman-Augmented State-Space Models

Master's Degree Thesis:
Hemodynamic Digital Twins: Fusing Hemodynamic Physics-Informed Neural Networks and Mamba-SSM for ICU Patient Prognostics

PROJECTS

Industrial Gas Turbine Digital Twin (Sonatrach) — 2026

- **Edge & Cloud Architecture**: Engineered a 3-tier framework decoupling safety-critical TCNEdge (Siemens PLC) from near-edge MambaTS and cloud-based ST-KDFormer.
- **Ultra-Low Latency Inference**: Compiled 1D Dilated Convolutions (TCNEdge) to INT8 ONNX C++, achieving 0.049ms latency for emergency shutdown.
- **Physcis & Robustness**: Implemented MambaTS with missingness-aware gating for broken sensors, and KAN-PCNN with PyTorch Autograd for absolute thermodynamic consistency.

Tech: MambaTS, TCNEdge, ST-KDFormer, KAN-PCNN, PyTorch, C++ ONNX, PINN

Cardiovascular Medical Digital Twin — 2026

- **High-Frequency Edge-Cloud Architecture**: Engineered **Medical_MambaTS** for sub-millisecond bedside inference and dual-branch **SST_KODA_MultiScale** for robust cloud forecasting using 100Hz VitalDB ICU telemetry.
- **Physics-Informed DeepONet**: Implemented a 0D Windkessel **PI_DeepONet** with PyTorch Autograd, forcing the model to obey cardiovascular fluid dynamics without retraining for new patients.
- **Data Pipeline & Robustness**: Developed a PyArrow Parquet zero-copy pipeline avoiding GPU starvation, and designed a custom missingness-aware gate to handle sensor dropouts natively.

Tech: PyTorch, Medical_MambaTS, PI_DeepONet, SST_KODA_MultiScale, Parquet PyArrow, Autograd

Aura — Enterprise AI Trading Platform — 2025

- **Hybrid Forecasting**: Built an autonomous trading system combining LSTM networks for time-series predictions.
- **Strategic Decision-Making**: Implemented Deep Q-Learning (Reinforcement Learning) to optimize long-term reward policies.
- **Real-Time Dashboard**: Engineered a high-throughput FastAPI backend with WebSockets and a Next.js interactive UI.

Tech: TensorFlow, PyTorch, FastAPI, Next.js, WebSockets, RL

GNN-based Recruiting System (Career Connect) — 2024–2025

- **Relational Learning**: Developed a recommendation engine using Graph Neural Networks (GNN) to map candidate-job relationships.
- **Semantic Embeddings**: Utilized PyTorch Geometric and SBERT for deep semantic text processing.
- **System Architecture**: Deployed via FastAPI and PostgreSQL, significantly outperforming keyword-based heuristics.

Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL, GNN

LANGUAGES

Arabic: Native | English: Advanced | French: Professional